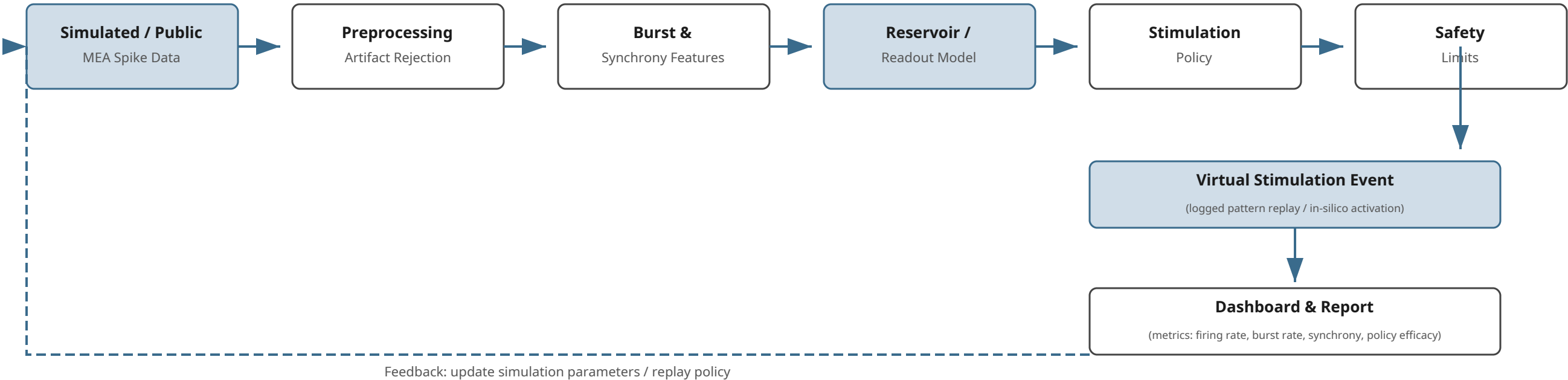


Feasible First Demonstration: Software-Only Closed Loop

End-to-end pipeline using simulated or public MEA data before wet-lab engagement



Data Sources	Preprocessing	Feature Extraction	Model	Stimulation Policy	Safety
SpikeInterface / MEA-NAP Neuropixels / public datasets Synthetic spike generators	Common-average reference Bandpass filter 300–3000 Hz Artifact template subtraction	Burst detection (ISI threshold) Synchrony (STTC, SPIKE-distance) Firing rate, CV, Fano factor	Echo State Network / LSM Ridge regression readout Cross-validated performance	Threshold / PID / RL policy Charge-balanced biphasic Channel selection logic	Max charge per phase: 4 nC Max frequency: 200 Hz Thermal / DC offset monitor

Example Run Log & Exit Criteria
Mean firing rate: 2.4 ± 0.8 Hz Burst rate: 0.8 ± 0.3 bursts/min STTC synchrony: 0.32 ± 0.06 Readout R²: 0.71 Policy hit rate: 78%
Exit: R² ≥ 0.7 AND policy hit rate ≥ 75% over 100 trials → ready for Phase 3 (commercial MEA adapter)
Otherwise: tune model hyperparameters (spectral radius, leaking rate) or increase simulation epochs → re-run